



• SOLAR ENERGY • LESSON 01

Introduction to Solar Energy

How the Sun Powers Buildings, Homes & Industry

Lesson 1 of 30 | Solar Energy Program | Beginner

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- Solar energy fundamentals — photons to electrons
- The photovoltaic effect — how panels produce power
- System architecture — panels, controller, battery, inverter
- Off-grid, on-grid, and hybrid system types compared
- Solar career pathways and global industry overview

01 | What Is Solar Energy?

Solar energy is the radiant light and heat produced by the sun, harnessed using technologies such as **photovoltaic (PV) panels**, solar thermal collectors, and concentrated solar power systems. The sun delivers more energy to Earth in one hour than all of humanity consumes in an entire year.

For engineers, solar energy is now the **cheapest source of electricity in history** — the cost of solar PV has dropped over 90% since 2010. This makes solar the fastest-growing energy technology worldwide, and solar engineering one of the most in-demand skills in the MEP and energy sectors.

SCALE — The Sun's Power Is Almost Incomprehensible

The sun radiates 3.8×10^{26} watts of power continuously. The solar irradiance reaching Earth's surface averages 1,000 W/m² on a clear day. A single square kilometre of desert receives enough solar energy per year to power a city of 500,000 people — with zero fuel cost and zero emissions.

02 | The Photovoltaic Effect — How Panels Generate Power

The photovoltaic effect was discovered by Edmond Becquerel in 1839. When photons (particles of light) strike a semiconductor material (typically silicon), they knock electrons loose. These free electrons flow through an external circuit, creating **direct current (DC) electricity**. This is the fundamental principle behind every solar panel in the world.

How Sunlight Becomes Usable Electricity:

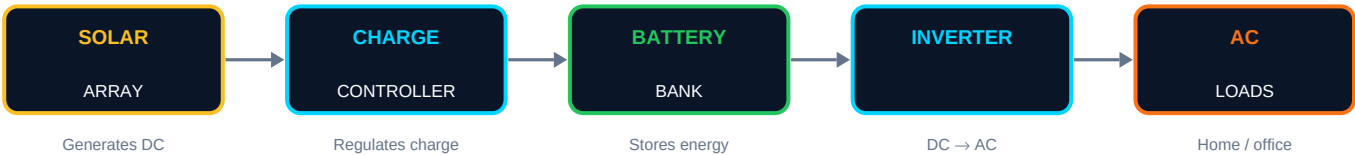


Level	Description	Output	Scale Example
Solar Cell	Single PV unit — silicon wafer	0.5 V / 3–4 W	Size of your palm
Solar Panel	60–72 cells wired in series/parallel	24–48 V / 250–450 W	1 m × 1.7 m sheet
Solar Array	Multiple panels wired together	Hundreds of volts / kW–MW	Rooftop, ground mount, solar farm
Solar Farm	Thousands of arrays at utility scale	Megawatts to Gigawatts	Covers km ² of land or desert

03 | Solar System Architecture — The 5 Core Components

Every solar power system — from a 3 kW home rooftop system to a 500 MW solar farm — is built from the same five fundamental components. Understanding each one and how they connect is the starting point of solar engineering.

Solar System — Component Flow



Component	Function	Key Spec to Know	Failure Effect
Solar Panels (PV)	Convert sunlight to DC electricity via the PV effect	Watt-peak (Wp), efficiency %, temperature coefficient	No power generation
Charge Controller	Regulates current from panels to battery — prevents overcharge	MPPT vs PWM type, voltage rating (V), current (A)	Battery damage or overcharge
Battery Bank	Stores DC electricity for use when sun is not available	Capacity (Ah / kWh), DoD %, cycle life	No power at night or cloudy days
Inverter	Converts DC from panels/battery to AC for household use	Power rating (W/kW), efficiency %, waveform type	No AC power output
Mounting & Wiring	Structural support and electrical connections for all components	Tilt angle (°), cable sizing (mm²), protection rating	Structural failure or fire risk

04 | Types of Solar Systems — Which One Goes Where

Not all solar systems work the same way. The three main types differ in how they handle energy storage and connection to the electrical grid. Choosing the right type is the first engineering decision in any solar project.



System Type	Grid Connection	Battery Storage	Best For	Main Advantage
Off-Grid	None — fully independent	Required (large bank)	Remote areas, farms, islands with no grid	Complete energy independence
On-Grid (Grid-Tie)	Yes — exports excess power	Not required	Urban homes and businesses with reliable grid	Lowest cost — no batteries needed
Hybrid	Yes — grid as backup	Required (smaller bank)	Areas with grid but frequent power cuts	Backup power + grid savings
Solar Water Pumping	None	Not required — pump directly	Agriculture, rural water supply	Simple, low-cost, no battery needed

05 | Solar Energy by the Numbers — What Every Engineer Must Know

Parameter	Typical Value	Why It Matters to Engineers
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Peak Sun Hours (PSH)	4–6 hrs/day (tropical)	Used to size panels: $\text{Energy needed} \div \text{PSH} = \text{panel kW}$
Panel Efficiency	18–23% (monocrystalline)	Higher efficiency = smaller area needed for same power
System Losses	20–25% total	Cable, inverter, temperature, dust — always factor in
Battery Depth of Discharge	80% (lithium) / 50% (lead-acid)	Usable capacity = rated Ah $\times$ DoD
Inverter Efficiency	93–98%	Multiply DC output by inverter efficiency for AC output
Payback Period	3–7 years (residential)	After payback, electricity is effectively free for 20+ yrs
Panel Lifespan	25–30 years	Panels degrade $\sim 0.5\%$ /year — factor into long-term yield
CO ₂ Savings	0.5–1 tonne CO ₂ /kWp/yr	Used in green building certifications and ESG reports

06 | Solar Career Pathways

Role	Core Responsibilities	Key Skills	Avg Salary (USD/yr)
Solar Design Engineer	System sizing, layout, energy yield reports	PVsyst, AutoCAD, load calc, shading analysis	\$55,000 – \$90,000
Solar Site Engineer	Supervise installation, testing, commissioning	Electrical safety, wiring, inverter setup	\$45,000 – \$75,000
Solar Project Manager	Manage budget, schedule, subcontractors	MS Project, cost control, procurement	\$70,000 – \$110,000
Solar Energy Auditor	Assess existing systems, identify efficiency gains	Energy monitoring, data analysis, BMS	\$55,000 – \$85,000
Solar BIM Specialist	Model solar systems in Revit, coordinate with MEP	Revit, Navisworks, IFC coordination	\$60,000 – \$95,000

Reflection:

"Look at the roof of your home, your office, or any building you pass today. Estimate the available roof area. If 1 kWp of solar requires roughly 6 m² of panels, how many kWp could that roof generate? What would that mean for the electricity bill?"

Action checklist:

Next up — SOLAR-002:

Solar Radiation Basics — peak sun hours, irradiance maps, tiltt.me/ayetechub angle optimisation, and shading analysis.

Community & support: